Python Programming

Assignment – 1

Name : Joshita Kunda

USN : 25BTRCC031

CSE-CYS

Problem 1 – Smart Shopping Bill Generator

Problem Understanding: A supermarket wants to automate its billing process. The program accepts the customer's name and the names/prices of three products, calculates the total, applies a 10% discount if the total exceeds ₹3000, and displays a full bill summary.

Algorithm:

Step 1 : Start

Step 2 : Read the customer name

Step 3 : For each of the 3 products, read the product name and its price

Step 4 : Compute Total Amount = sum of the 3 product prices

Step 5 : If Total Amount > 3000, then

Discount = 10% of Total Amount

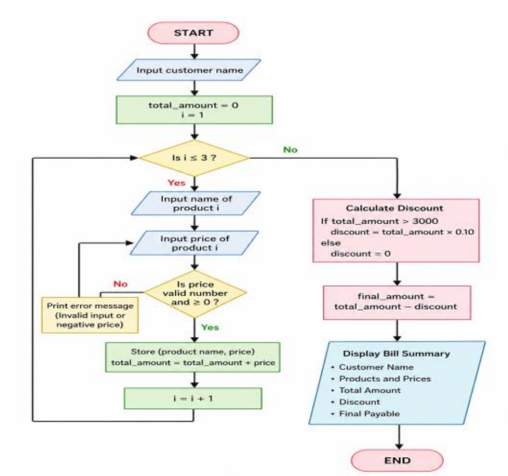
Else Discount = 0

Step 6 : Compute Final Amount = Total Amount - Discount

Step 7 : Display Customer Name, Product Details, Total Amount, Discount and Final Amount

Step 8 : Stop

Flowchart :



Python Source Code:

```
def generate_bill(): DISCOUNT_THRESHOLD = 3000 DISCOUNT_RATE = 0.10

customer_name = input("Enter customer name: ")

total_amount = 0.0
products = []

for i in range(1, 4):
   name = input(f"Enter name of product {i}: ")

   while True:
      try:
         price = float(input(f"Enter price of {name} (Rs.): "))
         if price >= 0:
            break
         print("Price cannot be negative.")
      except ValueError:
         print("Invalid price.")

   total_amount += price
   products.append((name, price))

discount = total_amount * DISCOUNT_RATE if total_amount > DISCOUNT_THRESHOLD else 0.0
final_amount = total_amount - discount

print("\n" + "=" * 40)
print("         SMART SHOPPING BILL")
print("=" * 40)
print(f"Customer Name : {customer_name}")
print("-" * 40)
print(f"{'Product':<20}{'Price (Rs.)':>20}")
print("-" * 40)

for name, price in products:
   print(f"{name:<20}{price:>20.2f}")
```

print("-" * 40)

print(f"{'Total Amount':<20}{total_amount:>20.2f}")

print(f"{'Discount':<20}{discount:>20.2f}")

print(f"{'Final Payable':<20}{final_amount:>20.2f}")

print("=" * 40)

if **name** == "**main**": generate_bill()

Output 1 :

```
PS C:\Users\Administrator\OneDrive\Docume
Enter customer name: Joshita
Enter name of product 1: shampoo
Enter price of shampoo (Rs.): 399
Enter name of product 2: Facewash
Enter price of Facewash (Rs.): 239
Enter name of product 3: Hairoil
Enter price of Hairoil (Rs.): 199

========================================
           SMART SHOPPING BILL
========================================
Customer Name : Joshita
----------------------------------------
Product                      Price (Rs.)
----------------------------------------
shampoo                           399.00
Facewash                          239.00
Hairoil                           199.00
----------------------------------------
Total Amount                      837.00
Discount                            0.00
Final Payable                     837.00
========================================
```

Output 2 :

```
PS C:\Users\Administrator\OneDrive\Documents\Py
Enter customer name: Dhyana
Enter name of product 1: Dark Fantasy Fills
Enter price of Dark Fantasy Fills (Rs.): 250
Enter name of product 2: Bingo Spicy Chips
Enter price of Bingo Spicy Chips (Rs.): 50
Enter name of product 3: Nescafe Coffee Powder
Enter price of Nescafe Coffee Powder (Rs.): 390

========================================
           SMART SHOPPING BILL
========================================
Customer Name : Dhyana
----------------------------------------
Product                      Price (Rs.)
----------------------------------------
Dark Fantasy Fills                250.00
Bingo Spicy Chips                  50.00
Nescafe Coffee Powder              390.00
----------------------------------------
Total Amount                      690.00
Discount                            0.00
Final Payable                     690.00
========================================
```

Output 3 :

```
Enter customer name: Dheemanth
Enter name of product 1: Battery
Enter price of Battery (Rs.): 39
Enter name of product 2: wire extension
Enter price of wire extension (Rs.): 599
Enter name of product 3: Sticky notes
Enter price of Sticky notes (Rs.): 99

========================================
           SMART SHOPPING BILL
========================================
Customer Name : Dheemanth
----------------------------------------
Product                      Price (Rs.)
----------------------------------------
Battery                            39.00
wire extension                    599.00
Sticky notes                       99.00
----------------------------------------
Total Amount                      737.00
Discount                            0.00
Final Payable                     737.00
========================================
```

Conclusion: The program correctly automates billing, applying the discount rule only when the total exceeds ₹3000, and was verified against below-threshold, above-threshold, and boundary test cases.

Problem 2 – Student Academic Performance Analyzer

Problem Understanding: Reads a student's name, USN, and marks in five subjects; computes total, percentage, average; assigns a grade based on percentage ranges; and prints a formatted report card.

Algorithm:

Step 1 : Start

Step 2 : Read Student Name and USN

Step 3 : Read marks obtained in 5 subjects (m1, m2, m3, m4, m5)

Step 4 : Compute Total Marks = m1 + m2 + m3 + m4 + m5

Step 5 : Compute Percentage = (Total Marks / 500) * 100

Step 6 : Compute Average = Total Marks / 5

Step 7 : Determine Grade based on Percentage:

if Percentage >= 90 -> Grade = "O"

else if Percentage >= 80 -> Grade = "A+"

else if Percentage >= 70 -> Grade = "A"

else if Percentage >= 60 -> Grade = "B"

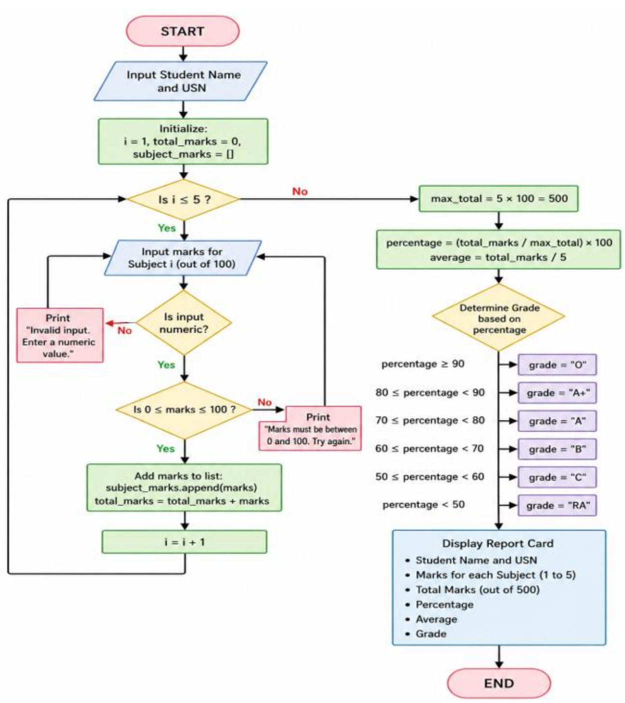
else if Percentage >= 50 -> Grade = "C"

else -> Grade = "RA"

Step 8 : Display Name, USN, Total Marks, Percentage, Average and Grade

Step 9 : Stop

Flowchart :



Program source code :

NUMBER_OF_SUBJECTS = 5 MAX_MARKS_PER_SUBJECT = 100

def analyze_performance():

student_name = input("Enter Student Name: ") usn = input("Enter USN: ")

```
subject_marks = []
total_marks = 0.0
for i in range(1, NUMBER_OF_SUBJECTS + 1):
    while True:
        try:
            marks = float(input(f"Enter marks in Subject {i} (out of 100):
"))
            if 0 <= marks <= MAX_MARKS_PER_SUBJECT:
                break
          print("Marks must be between 0 and 100.")
        except ValueError:
            print("Invalid input.")

    subject_marks.append(marks)
    total_marks += marks

percentage = (total_marks / (NUMBER_OF_SUBJECTS * MAX_MARKS_PER_SUBJECT)) *
100
average = total_marks / NUMBER_OF_SUBJECTS

if percentage >= 90:
    grade = "O"
elif percentage >= 80:
    grade = "A+"
elif percentage >= 70:
    grade = "A"
elif percentage >= 60:
    grade = "B"
elif percentage >= 50:
    grade = "C"
else:
    grade = "RA"
```

```
print("\n" + "=" * 42)
print("        STUDENT ACADEMIC REPORT CARD")
print("=" * 42)
print(f"Student Name : {student_name}")
print(f"USN          : {usn}")
print("-" * 42)

for i, marks in enumerate(subject_marks, start=1):
    print(f"Subject {i} Marks : {marks:.2f} / {MAX_MARKS_PER_SUBJECT}")

print("-" * 42)
print(f"Total Marks : {total_marks:.2f} / {NUMBER_OF_SUBJECTS *
MAX_MARKS_PER_SUBJECT}")
print(f"Percentage  : {percentage:.2f}%")
print(f"Average     : {average:.2f}")
print(f"Grade       : {grade}")
print("=" * 42)
```

if **name** == "**main**": analyze_performance()

Output 1 :

```
Enter Student Name: Joshita
Enter USN: 25BTRCC031
Enter marks in Subject 1 (out of 100): 87
Enter marks in Subject 2 (out of 100): 90
Enter marks in Subject 3 (out of 100): 85
Enter marks in Subject 4 (out of 100): 79
Enter marks in Subject 5 (out of 100): 91

==========================================
        STUDENT ACADEMIC REPORT CARD
==========================================
Student Name : Joshita
USN          : 25BTRCC031
------------------------------------------
Subject 1 Marks : 87.00 / 100
Subject 2 Marks : 90.00 / 100
Subject 3 Marks : 85.00 / 100
Subject 4 Marks : 79.00 / 100
Subject 5 Marks : 91.00 / 100
------------------------------------------
Total Marks : 432.00 / 500
Percentage  : 86.40%
Average     : 86.40
Grade       : A+
==========================================
```

Output 2 :

```
Enter Student Name: Bhanupriya
Enter USN: 25BTRCC028
Enter marks in Subject 1 (out of 100): 88
Enter marks in Subject 2 (out of 100): 78
Enter marks in Subject 3 (out of 100): 67
Enter marks in Subject 4 (out of 100): 83
Enter marks in Subject 5 (out of 100): 75

==========================================
        STUDENT ACADEMIC REPORT CARD
==========================================
Student Name : Bhanupriya
USN          : 25BTRCC028
------------------------------------------
Subject 1 Marks : 88.00 / 100
Subject 2 Marks : 78.00 / 100
Subject 3 Marks : 67.00 / 100
Subject 4 Marks : 83.00 / 100
Subject 5 Marks : 75.00 / 100
------------------------------------------
Total Marks : 391.00 / 500
Percentage  : 78.20%
Average     : 78.20
Grade       : A
==========================================
```

Output 3 :

```
Enter Student Name: Shagufta
Enter USN: 25BTRCC050
Enter marks in Subject 1 (out of 100): 86
Enter marks in Subject 2 (out of 100): 78
Enter marks in Subject 3 (out of 100): 90
Enter marks in Subject 4 (out of 100): 81
Enter marks in Subject 5 (out of 100): 92

==========================================
        STUDENT ACADEMIC REPORT CARD
==========================================
Student Name : Shagufta
USN          : 25BTRCC050
------------------------------------------
Subject 1 Marks : 86.00 / 100
Subject 2 Marks : 78.00 / 100
Subject 3 Marks : 90.00 / 100
Subject 4 Marks : 81.00 / 100
Subject 5 Marks : 92.00 / 100
------------------------------------------
Total Marks : 427.00 / 500
Percentage  : 85.40%
Average     : 85.40
Grade       : A+
==========================================
```

Conclusion: The analyzer correctly computes total, percentage, average and grade, verified across excellent, average, and failing score ranges.

Problem 3 – Personal Utility Toolkit

Problem Understanding: A menu-driven application offering Celsius↔Fahrenheit conversion, area of a circle, and simple interest calculation, looping until the user exits.

Algorithm:

Step 1 : Start

Step 2 : Repeat Steps 3 to 5 until the user chooses to Exit

Step 3 : Display the menu with options 1 to 5

Step 4 : Read the user's choice

Step 5 : Based on the choice:

1 -> Convert Celsius to Fahrenheit and display result

2 -> Convert Fahrenheit to Celsius and display result

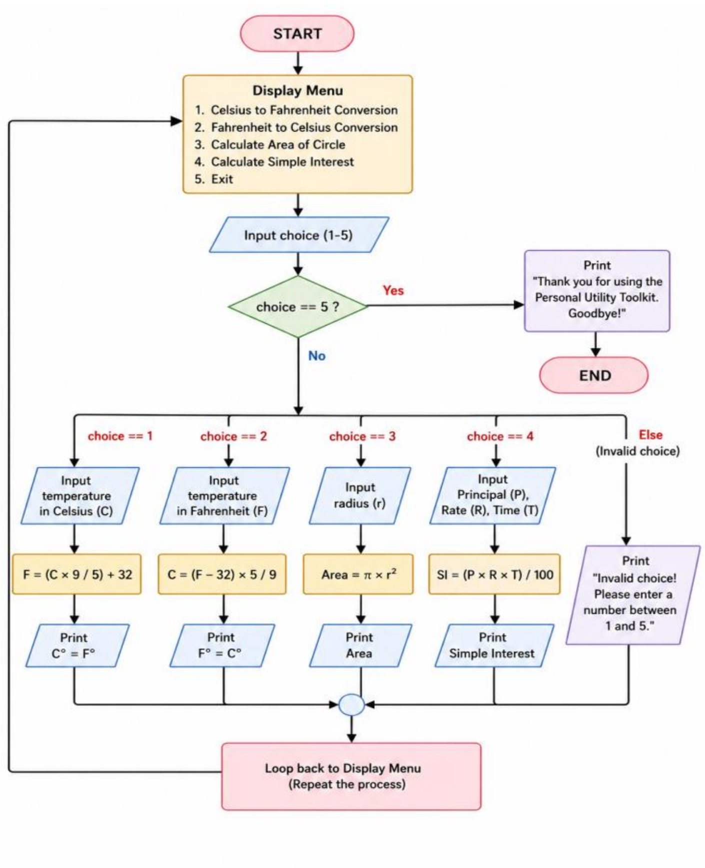
3 -> Read radius, compute Area = pi * r^2, display result

4 -> Read principal, rate, time; compute Simple Interest = (P * R * T) / 100, display result

5 -> Exit the program other -> Display "Invalid choice" message

Step 6 : Stop

Flowchart :



Output 1 :

```
========== PERSONAL UTILITY TOOL ==========
1. Celsius to Fahrenheit Conversion
2. Fahrenheit to Celsius Conversion
3. Calculate Area of Circle
4. Calculate Simple Interest
5. Exit
===========================================
Enter your choice (1-5): 4
Enter Principal Amount (Rs.): 11500
Enter Rate of Interest (%): 0.75
Enter Time (Years): 1
Simple Interest = Rs. 86.25
```

Output 2 :

```
========== PERSONAL UTILITY TOOL ==========
1. Celsius to Fahrenheit Conversion
2. Fahrenheit to Celsius Conversion
3. Calculate Area of Circle
4. Calculate Simple Interest
5. Exit
===========================================
Enter your choice (1-5): 1
Enter temperature in Celsius: 89
89.0°C = 192.20°F
```

Output 3 :

```
========== PERSONAL UTILITY TOOL ==========
1. Celsius to Fahrenheit Conversion
2. Fahrenheit to Celsius Conversion
3. Calculate Area of Circle
4. Calculate Simple Interest
5. Exit
===========================================
Enter your choice (1-5): 3
Enter radius of the circle: 30
Area of Circle = 2827.43 sq. units
```

Output 4 :

```
========== PERSONAL UTILITY TOOL ==========
1. Celsius to Fahrenheit Conversion
2. Fahrenheit to Celsius Conversion
3. Calculate Area of Circle
4. Calculate Simple Interest
5. Exit
===========================================
Enter your choice (1-5): 2
Enter temperature in Fahrenheit: 93
93.0°F = 33.89°C
```

Conclusion: The toolkit successfully implements a menu-driven loop offering four independent utilities, handles invalid input gracefully, and exits cleanly on request.

Github Repository Link : https://github.com/juug25btech20073-crypto/Python-Programming_25BTRCC031